Evidence-blind confidence in a streaming video-LLM

E2 headline findings — Qwen3-VL-8B + EgoStream on EgoLifeQA

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373 questions · 17,311 streaming records · frozen Qwen3-VL-8B-Instruct

Agenda

1. **Setup and expectation.**
   Can model "confidence" serve as a proxy for metamemory?
2. **The result: confidence is evidence-blind.**
   At every Δt where the no-evidence floor was measured, accuracy responds and confidence does not.
3. **What confidence actually tracks.**
   The context-volume dose–response: a monotone on one measure, flat on the other.
4. **Off-letter mass: an implicit abstention signal.**
   When evidence is gone, the model sometimes stops wanting to pick a letter.
5. **AUROC degradation: a genuinely new phenomenon.**
   The confidence *ranking* decays under a bounded cache, where it was flat before.

Setup and expectation

A streaming KV cache that genuinely evicts evidence

The system. Qwen3-VL-8B processes egocentric video through EgoStream's bounded KV cache. As new frames arrive, old tokens are pruned. Evidence is genuinely destroyed: 0 of 13,636 default/large rows offloaded anything.

The probe (Tsvilodub et al. 2024 / EgoSAT Eq. 6):

- **Accuracy**: argmax over {A,B,C,D} vs gold.
- **Confidence**: max $\mathrm{softmax}(\{\mathrm{A,B,C,D}\})$ at the first generated position.

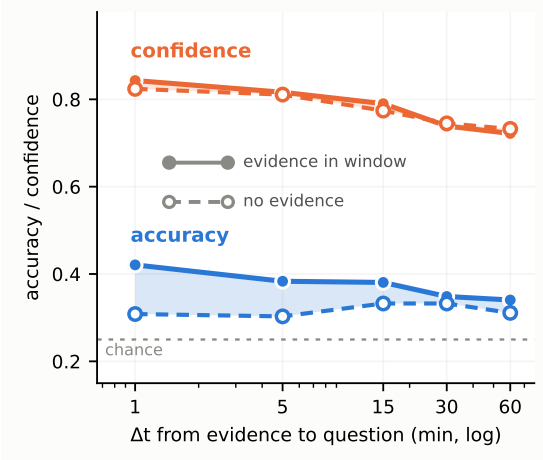
The key manipulation, paired within each of 373 questions at $\Delta t \in \{1, 5, 15, 30 \text{ and } 60\}$ min:

`main@`Δt	`prior@`Δt
window includes the evidence frames	window of *same* duration *before* the evidence

Expectation. If "confidence" proxies metamemory, removing the evidence should lower it in proportion to the accuracy loss.

Result: confidence is evidence-blind

The largest accuracy effect in E2 is invisible to confidence — in every bin



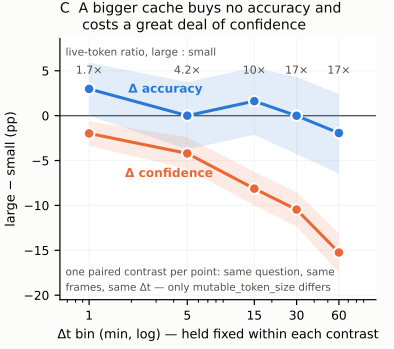
Token-matched, paired within each of 373 questions:

Δt (min)	Δaccuracy (pp)	Δconfidence (pp)
1	+11.3 [+6.2, +16.6]	+1.9 [−0.2, +4.1]
5	+8.1 [+2.9, +13.1]	+0.5 [−1.5, +2.6]
15	+4.9 [+0.3, +9.4]	+1.6 [−0.4, +3.5]
30	+1.6 [−2.4, +5.6]	−0.6 [−2.4, +1.2]
60	+2.9 [−1.1, +7.0]	−1.1 [−3.1, +0.9]

Δaccuracy runs +1.6 to +11.3 pp, excluding zero in 3 of 5 bins — the advantage decays as the cache saturates. **Δconfidence** spans zero in all 5 and tracks neither the effect nor its decay.

What confidence actually tracks

Context volume — not evidence, not elapsed time



Budget dose–response (`large − small`) at each Δt bin.

Varying the memory budget 17× at $\Delta t = 60$ min:

	Δ (pp)
Δaccuracy	−1.9 [−6.4, +2.4]
Δconfidence	−15.2 [−17.5, −13.1]
offload-free pair (3.7×):	
Δaccuracy	*null in every bin*
Δconfidence	−6.0 [−7.3, −4.7]

Decomposition of $\Delta t = 15 \rightarrow 60$ min (residual +0.02 pp):

term	acc	conf
elapsed time	−1.68	−0.13
context	−2.39	−6.71

More context buys no accuracy and costs a great deal of confidence. The sweep's confidence decline is **entirely** the context term (−6.71 of −6.82 pp).

Off-letter mass: an implicit abstention signal

When the evidence is gone, the model sometimes stops wanting to pick a letter

The restricted softmax discards mass the model places off {A,B,C,D}. In **E1** that mass is negligible (max 0.0034). In **E2**: 130 rows (1.29%), over 79 of 373 questions put >10 % off the letters — 5.31% of the `prior` arm vs 0.42% of `main`.

Paired within-question contrast on off-letter mass:

contrast	Δ off-letter (pp)
evidence removed	−3.01 [−4.40, −1.82]
raw Δt sweep (1→60 min)	+0.60 [+0.21, +1.11]

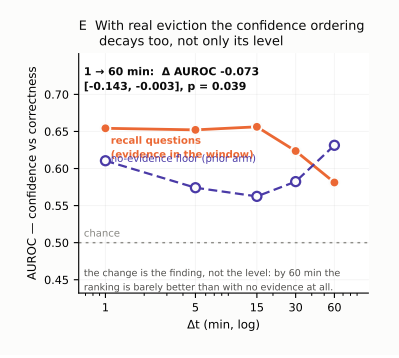
Removing the evidence **raises off-letter mass by 3.0 pp** ($p < 0.001$). An evidence-sensitive signal — an **implicit abstention** — in exactly the regime where max-p is silent.

Exists **only under the bounded cache**: **E1**'s equivalent is three orders of magnitude smaller. Free from committed records.

Source: `memo/CONFIDENCE-PROBE.md` §5.

AUROC degradation: the one genuinely new phenomenon

The confidence ranking decays when evidence is genuinely evicted



AUROC of confidence against correctness over the Δt sweep.

AUROC = how well confidence *ranks* correct above incorrect answers (1.0 = perfect separation, 0.5 = chance). Invariant to confidence level — a pure ranking measure.

AUROC change, Δt $1 \rightarrow 60$ min:

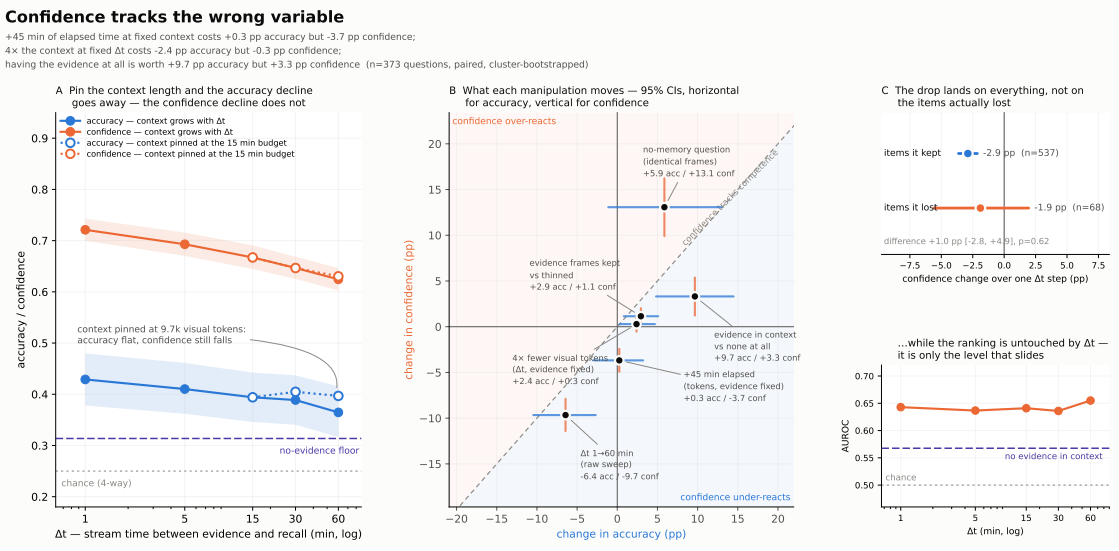
E1	$+0.012$ $[-0.046, +0.071]$
E2	-0.073 $[-0.143, -0.003]$
	$p = 0.039$

Under full-reprocess, evidence is diluted but never removed — the ranking stays flat. Under a bounded cache, evidence is *genuinely evicted* — and the ranking decays with it.

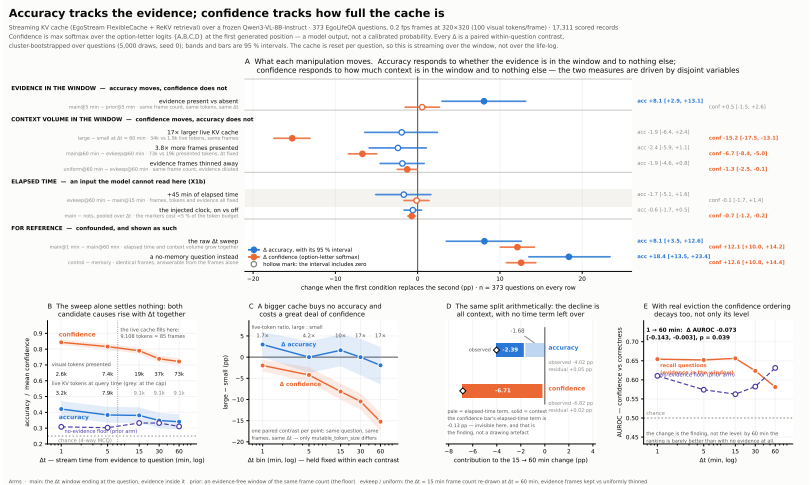
Per-question confidence is a poor proxy for metamemory (P2–P3). But population-level AUROC *is* sensitive to the model's memory state — it degrades exactly when evidence is evicted. It may serve as an evaluation metric for metamemory even when the raw score does not.

Source: `artifacts/x9_streaming.md`, discrimination section.

Backup — E1 headline figure (full-reprocess)



Backup — E2 headline figure (streaming KV cache)



Backup — confidence probe definition and robustness

Definition. $\text{confidence} = \max_{L \in \{A,B,C,D\}} \text{softmax}(z_L)$ at the first generated position, no text generated. Correctness is argmax vs gold. This is Tsvilodub et al. 2024 / EgoSAT Eq. 6.

Denominator. Restricted to the four option letters. Mean mass on {A,B,C,D}: 0.99988 (**E1**), 0.99546 (**E2**). Argmax disagreements: **0 of 24,433** records.

Denominator sensitivity. Vocabulary-normalised softmax is inert except on **E2** `evidence_vs_none`, where it gives **+2.61 [+0.33, +4.94]** — driven by the off-letter tail (P4).

Robustness. AUROC is invariant to scalarisation:

discriminator	E1	E2
max-p, restricted	0.6276	0.6196
max-p, vocab-normalised	0.6275	0.6198
negative entropy	0.6228	0.6209
margin (top1 − top2)	0.6254	0.6169

Option-letter bias. E1 predicts C on 43.5 %, **E2** predicts B on 34.2 %, against near-uniform gold. Paired design cancels it. EgoSAT runs the cyclic-permutation check we do not.

Source: `memo/CONFIDENCE-PROBE.md`

Backup — dataset curation: 500 → 373 questions

step	in	out	reason
EgoLifeQA A1 (full)	500	—	
Drop TaskMaster	500	437	prospective
Screen HabitInsight	437	376	no single anchor
Evidence span >15 min	376	373	ambiguous Δt

EntityLog / EventRecall / RelationMap at 125 / 124 / 124. All 373 in every Δt bin and arm. Token-matched arms hold visual tokens exactly at the $\Delta t = 15$ min level.

TaskMaster (63): prospective, incompatible with retrospective Δt. 93.7 % require audio.

HabitInsight (61): no single evidence event to anchor Δt. Scored separately.

Span exclusions (3): q482 spans 50.7 h. q189/q346 at 19/21 min.

EgoLifeQA participant A1/Jake — the only released participant. One model, one cache policy.

Source: M1 `m1_questions.py`, M4 `m4_plan.py`.

Backup — self-use: what is *not* in these slides

1. **Cross-regime rotation.** Confidence driver swaps: elapsed time in **E1** (**−3.7 [−4.9, −2.4]**), context volume in **E2** (**−6.7 [−8.4, −5.0]**). Differences exclude zero (**+3.5 [+1.7, +5.4]** and **−6.1 [−8.1, −4.1]**). On evidence: **E1** gave **+9.7 [+4.8, +14.5]**/**+3.3 [+1.2, +5.4]**; cross-regime Δconfidence **−2.8 [−5.1, −0.4]**.
2. **Elapsed time is *unavailable* in E2, not null.** EgoStream's image path has no native time markers; injected ones are unreadable once the cache saturates.
3. **Item-level metamemory.** Suggestive, underpowered: −3.8 [−8.5, +1.1], $p = 0.118$, on 107 lost items.
4. **Original hypothesis rejected.** Widening gap absent both times (**E1**: −3.2 [−7.5, +1.0], $p = 0.138$; **E2**: −4.0 [−9.0, +1.0], $p = 0.105$).
5. **Cheapest follow-ups.** (a) Strip timestamps from **E1** (≈1 GPU-h). (b) Recalibrate on `live_tokens` (free). (c) Top-k dump on `prior` for off-letter mass.

Backup — scope conditions

- **Window, not life-log.** Each question resets the cache. No claim about 44 h of history.
- **Evidence effect resolved at $\Delta t \leq$ 15 min only.** The floor is measured at every bin; by 30–60 min the contrast itself is within noise.
- **Levels not comparable across regimes. E2** changes prompt and ingestion path. Contrasts are comparable.
- **Dynamic range $\sim$11 pp.** Bin-1 accuracy 42.1 % against floor 30.3 %.
- **One participant, one model, one cache policy.**
- **"Confidence" is not a calibrated probability.** Report as "option-letter softmax".